

Motion-Painter Canonical Guide — Updated (Ambigram | packingSDF | Minimizer Integration)

Integrated roadmap, staged tasks, APIs, and contracts for safe, incremental rollout

Prepared by ChatGPT (assistant) — 2025-10-04

Motion-Painter Canonical Guide (original): [■filecite■turn2file0■](#)

Ambigram■anamorph Module — Specification: [■filecite■turn2file1■](#)

Packing Sdf Module Spec: [■filecite■turn2file2■](#)

Minimizer Module — Algorithm Guide: [■filecite■turn2file3■](#)

Executive summary

Executive summary This updated canonical guide folds the following new workflows into the Motion-Painter roadmap: 1) Early camera metrics & keypoint extraction as first-class data objects. 2) Canonical container construction from camera metrics (frustums, keypoint hulls) and passing these containers to the Minimizer module. 3) Fresnel-annular eviction output computed in the eviction hook (upstream of packingSDF). 4) packingSDF streaming & motion-warm-start integration producing signed SDF / narrowMask / disks for downstream modules. 5) PascalQuad / PSQ encoding + AmbiAdapter that consumes PSQ, disks and keypoint heatmaps to produce optimized ambigram geometry and LUTs. The new flow is designed to be incremental, workerized, feature-flagged, and fallback-safe.

Updated architecture & dataflow (high-level)

Key additions: - CameraMetrics & KeypointWorker: run immediately after frame eviction. Produce camera intrinsics/extrinsics, poseHash, and keypoint 2D/3D + gaussian heatmaps. - EvictionHook Fresnel step: compute hybrid Fresnel-annular artifact that seeds packingSDF densityMap. - PackingSDF (streaming): compute/update signed_sdf, narrowMask, disks[] using motion-warp warm starts. Persist artifacts to indexed storage. - ContainerBuilder: build Minimizer canonical containers (cameraFrustum, frustumDepthSlice, keypointHull) from camera metrics and keypoints. - Minimizer (canonical domain): setDomain(container) → buildVoxelSDF → solveMinimizer → export narrow SDF/mesh. Persist per pose-bucket. - PSQWorker (PascalQuad): encode SDF/primitives; produce seeds for AmbiAdapter. - AmbiAdapter: consumes PSQ, disks, keypoint heatmaps, and camera poses to optimize ambigram slats and generate EXR LUTs for runtime. - CompositeRenderer: runtime LUT application + optional shader-nudge using keypoint deltas. This flow keeps heavy optimization offline or in background workers and uses narrow-band streaming for runtime responsiveness. References: original Motion-Painter guide and module specs. See citations at the front.

Staged plan — ordered implementation

Stage 0 — Preparation (Feature flags, storage schema, telemetry)

- Add feature flags in app.config (enableFresnelEviction, enablePackingSdf, enableKeypointPipeline, enableAmbiAdapter, enablePascalQuadSdf).
- Add versioned storage keys and schema migration.
- Add telemetry counters and backpressure rules (workerQueue caps).

Stage 1 — Camera metrics & Keypoints (early)

- Implement KeypointWorker (recommended: TensorFlow.js pose-detection or MediaPipe) inside PreprocessorWorker. Emit keypoints2D, confidence, and gaussian heatmaps.
- Record camera intrinsics/extrinsics per frame; compute poseHash.
- Persist artifacts: mp:artifact:kp:{frameKey}:v1 and camera metadata.

Stage 2 — Fresnel annular output (in eviction hook)

- Compute hybrid Fresnel-annular artifact in eviction hook (workerized).
- Persist as mp:artifact:fresnel:{frameKey}:v1 and provide as densityMap input to packingSDF.

Stage 3 — packingSDF streaming integration

- Call PackingSdf.computeAll / updateFrame after Fresnel. Use motion-warp warm starts and delta-EDT.
- Return signed_sdf, narrowMask, disks, normals, densityMap; persist mp:artifact:packingSdf:{frameKey}:v1.
- Expose updateRegion to bound downstream recompute.

Stage 4 — PascalQuad (PSQ) encoding + AmbiAdapter

- PSQWorker encodes compact primitives from narrow-band SDFs and disks.
- AmbiAdapter builds representation using PSQ primitives, disks and keypoint heatmaps, then runs narrow-band local optimizers to produce LUTs and slats.
- Full optimizers run offline or when workerQueue permits.

Stage 5 — Minimizer canonical containers & solving (inserted early)

- ContainerBuilder constructs canonical domain descriptors from camera metrics: cameraFrustum, frustumDepthSlice, keypointHull.
- Pass container to Minimizer.setDomain(container) and run buildVoxelSDF(s) and solveMinimizer for pose-buckets.
- Persist minimizer artifacts per poseHash for LUT reuse.

Stage 6 — Runtime application & UX

- CompositeRenderer applies EXR LUTs; if exact LUT missing use nearest/interpolated pose-bucket then apply small shader-nudge driven by keypoint deltas.
- Add debug overlays: signed_sdf, narrowMask, disks, keypoint heatmap, ambiLUT.

Stage 7 — Tests, rollout, and monitoring

- Unit tests: packingSDF EDT/packing, Minimizer analytic cases, AmbiAdapter local optimization.
- Integration tests: two-view ambigram IoU threshold, long-run stability, workerQueue caps.
- Canary rollout behind feature flags.

API & message contracts (examples)

Examples (PostMessage / Storage keys):

PreprocessorWorker -> Manager (packingSdf result)

```
{
  "type": "packingSdfResult",
  "frameKey": "",
  "gridSize": [1024, 1024],
  "signedSdf": "",
  "narrowMask": "",
  "disks": [{"x": 123, "y": 456, "r": 6.2}, ...],
  "meta": {"version": "v1", "approx": false}
}
```

Keypoint result

```
{
  "type": "kpResult",
  "frameKey": "",
  "keypoints2D": [{"id": 0, "x": 123, "y": 456, "score": 0.92}, ...],
  "heatmapBuffer": ""
}
```

Ambi result

```
{
  "type": "ambiResult",
  "poseHash": "",
  "lut": "",
  "qualityMetrics": {"iou": 0.92, "kpAlign": 0.87}
}
```

Minimizer container (canonical)

```
{
  "type": "canonical",
  "name": "cameraFrustum",
  "params": {
    "poseMatrix": [...16 floats...],
    "fovDeg": 45,
    "aspect": 1.78,
    "near": 0.1,
    "far": 3.0,
    "imageSize": [1920, 1080],
    "padding": 0.05
  }
}
```

Practical parameters & heuristics

- SDF raster size: 1024×1024 for crisp MSDF/text tasks; lower for faster runtime. - PackingSDF narrow band: 8–16 cells. - Minimizer voxel grid: interactive ~128^3, export high-fidelity 256^3+. - Pose quantization for LUTs: ~1° / 1cm buckets (tunable). - Disk radii rMin/rMax: start with rMin=0.005, rMax=0.1 (world units), adapt via densityMap. - Backpressure: pause non-critical work if workerQueue > 6.

Acceptance criteria (what 'done' looks like)

- Camera metrics and keypoint heatmaps produced reliably per frame and visible in debug panel. - Fresnel artifact emitted at eviction and seeding packingSDF. - packingSDF provides stable disks[] with low flicker under small motion. - Minimizer produces narrow-band SDFs and reusable minimizers for pose buckets. - AmbiAdapter can generate an LUT and CompositeRenderer applies it visually. - No regressions in core rendering framerate beyond acceptable tolerance and worker queue remains bounded.

References (source specifications used)

Motion-Painter Canonical Guide (original): ■filecite■turn2file0■

Ambigram■anamorph Module — Specification: ■filecite■turn2file1■

Packing Sdf Module Spec: ■filecite■turn2file2■

Minimizer Module — Algorithm Guide: ■filecite■turn2file3■

Generated by ChatGPT; includes integrated design choices from the Motion-Painter canonical guide and module specs. File artifacts referenced are included as citations for traceability.